Widgets & UI

Composing layouts and handling gestures

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Lab objectives

Compose a screen

Build a todo list from Column, Row, ListView and Card, without a single hard-coded pixel offset.

Handle a gesture

Swipe a row away with Dismissible and keep the list consistent.

Time
2 hours

Present
To the lab instructor, at the end of this lab

Project setup

1. Open the project from Lab 2.
2. Create `lib/lab3/todo_screen.dart`. Everything in this lab lives there.
3. Run the app once before changing anything. If it does not build, fix that first.

Hint

Hot reload keeps your list contents between edits. Hot restart drops them: use it after changing `initState`.

The todo screen

1. An AppBar titled *Todos*.
2. A ListView.builder with one Card per todo.
3. A Checkbox in each Card, bound to `done`.
4. A FloatingActionButton that appends a todo.
5. Extract the Card into `TodoTile`.

Done when

- [ ] Adding a todo scrolls the list, and the list does not jump.
- [ ] Toggling a checkbox rebuilds only that tile.
- [ ] `TodoTile` takes its data through the constructor.

Reusable component means constructor in, callbacks out. A tile that reaches into global state is not reusable.

A swipeable row

```
Dismissible(
  // Key by id, never by the index
  key: ValueKey(todo.id),
  direction: DismissDirection.endToStart,
  onDismissed: (_) => onDelete(todo.id),
  background: Container(color: Colors.red),
  child: TodoTile(
    todo: todo,
    onToggle: onToggle,
  ),
)
```

1. Wrap each tile in a Dismissible.
2. Remove the item from the list in `onDismissed`.
3. Show a SnackBar with an *Undo* action.

Watch out

Using the list index as the key makes the wrong row disappear after a delete.

Layout errors

`RenderFlex overflowed by 42 pixels`	A Row or Column has more content than space. Wrap the growing child in Expanded or Flexible.
`Vertical viewport was given unbounded height`	A ListView inside a Column. Give it a fixed height, or wrap it in Expanded.
`setState() called after dispose()`	An async callback finished after the screen was popped. Check `mounted` before calling setState.
`A dismissed Dismissible widget is still part of the tree`	You did not remove the item from the list in onDismissed.

Presentation

The todo screen running · swipe to dismiss, shown live